Example 1.19

Solve the equations $2x + 3y = 7$, $3x + 5y = 9$ by Cramer's rule.

Solution:

The equations are

$$2x + 3y = 7$$
$$3x + 5y = 9$$

Here $\Delta = \begin{vmatrix} 2 & 3 \\ 3 & 5 \end{vmatrix} = 1 \neq 0$

$\therefore$ we can apply Cramer's Rule

Now $\Delta_x = \begin{vmatrix} 7 & 3 \\ 9 & 5 \end{vmatrix} = 8$ $\Delta_y = \begin{vmatrix} 2 & 7 \\ 3 & 9 \end{vmatrix} = -3$

$\therefore$ By Cramer's rule

$$x = \frac{\Delta x}{\Delta} = \frac{8}{1} = 8 \quad y = \frac{\Delta y}{\Delta} = \frac{-3}{1} = -3$$

$\therefore$ Solution is $x = 8, y = -3$

Example 1.20

The following table represents the number of shares of two companies A and B during the month of January and February and it also gives the amount in rupees invested by Ravi during these two months for the purchase of shares of two companies. Find the the price per share of A and B purchased during both the months.

Months	Number of Shares of the company		Amount invested by Ravi (in ₹)
	A	B	
January	10	5	125
February	9	12	150

Solution:

Let the price of one share of A be x

Let the price of one share of B be y

$\therefore$ By given data, we get the following equations

$$10x + 5y = 125$$
$$9x + 12y = 150$$

$$\Delta = \begin{vmatrix} 10 & 5 \\ 9 & 12 \end{vmatrix} = 75 \neq 0$$

$$\Delta_x = \begin{vmatrix} 125 & 5 \\ 150 & 12 \end{vmatrix} = 750 \quad \Delta_y = \begin{vmatrix} 10 & 125 \\ 9 & 150 \end{vmatrix} = 375$$

$\therefore$ By Cramer's rule

$$x = \frac{\Delta x}{\Delta} = \frac{750}{75} = 10 \quad y = \frac{\Delta y}{\Delta} = \frac{375}{75} = 5$$

The price of the share A is ₹10 and the price of the share B is ₹5.

Example 1.21

The total cost of 11 pencils and 3 erasers is ₹ 64 and the total cost of 8 pencils and 3 erasers is ₹49. Find the cost of each pencil and each eraser by Cramer's rule.

Solution:

Let 'x' be the cost of a pencil

Let 'y' be the cost of an eraser

$\therefore$ By given data, we get the following equations

$$11x + 3y = 64$$
$$8x + 3y = 49$$

$\Delta = \begin{vmatrix} 11 & 3 \\ 8 & 3 \end{vmatrix} = 9 \neq 0$. It has unique solution.

$$\Delta_x = \begin{vmatrix} 64 & 3 \\ 49 & 3 \end{vmatrix} = 45 \quad \Delta_y = \begin{vmatrix} 11 & 64 \\ 8 & 49 \end{vmatrix} = 27$$

$\therefore$ By Cramer's rule

$$x = \frac{\Delta x}{\Delta} = \frac{45}{9} = 5 \quad y = \frac{\Delta y}{\Delta} = \frac{27}{9} = 3$$

$\therefore$ The cost of a pencil is ₹ 5 and the cost of an eraser is ₹ 3.

Example 1.22

Solve by Cramer's rule $x + y + z = 4$, $2x - y + 3z = 1$, $3x + 2y - z = 1$

Solution:

Here $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{vmatrix} = 13 \neq 0$

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